

Leo Shaw

leos@andrew.cmu.edu

408.769.9693

[linkedin.com/in/leoshaw](https://www.linkedin.com/in/leoshaw)

github.com/leoshaw

Education

Carnegie Mellon University

Master of Science in Artificial Intelligence Engineering - Mechanical Engineering

Pittsburgh, PA

Dec 2026

Rutgers University

Bachelor of Science in Mechanical Engineering; Bachelor of Arts in Computer Science

New Brunswick, NJ

Dec 2024

Skills

Programming: Python, C++, Java, PyTorch, TensorFlow, Keras, OpenCV, YOLO, OCR, VLMs, RAG

Data/Infrastructure: PostgreSQL, Docker, FastAPI, JSONL

Design/Hardware: SOLIDWORKS, ANSYS, PLC/HMI integration, circuitry, soldering, 3D printing

Research & Projects

AI Research Engineer

Pittsburgh, PA / Carnegie Mellon University, May 2026 - Present

- Architected multimodal AI data pipelines for semiconductor and advanced manufacturing workflows, combining OCR, VLMs, structured extraction, and retrieval-oriented preprocessing for downstream model training.
- Built a Dockerized PostgreSQL backbone for a multi-billion-token manufacturing AI corpus, designing ingestion and storage workflows for pretraining, RAG, chain-of-thought fine-tuning, tree-of-thought traces, troubleshooting records, and process telemetry.
- Engineered a production-style OCR/VLM ingestion stack that transforms scientific papers, technical manuals, figures, tables, and scanned operator logs into database-ready records with layout-aware chunking, metadata tracking, quality scoring, and export pipelines.
- Designed separate pipelines for scientific documents and handwritten operator notes, improving generalization by pairing article-grade parsing with form-aware preprocessing, schema extraction, and equipment-log correlation.
- Built regression and evaluation tooling to benchmark ingestion quality, throughput, and database readiness across multiple document classes, supporting reliable iteration on AI data infrastructure.

Autonomous Driving Local Planner with Reinforcement Learning

Jan 2026 - May 2026

- Engineered an end-to-end autonomy pipeline in CARLA for highway merge and unprotected left-turn scenarios, combining graph-based routing, PPO policy learning, deterministic control, and route-conditioned state features in a closed-loop simulation stack.
- Built the full ML experimentation workflow in Python with PyTorch and CARLA, including scenario generation, reward shaping, training orchestration, rollout evaluation, and performance benchmarking.
- Improved policy performance across both driving tasks, increasing success rates from 76% to 83% and 44% to 80%, while reducing highway-merge 90th-percentile completion time from 24.5s to 15.25s.
- Structured the project as a reusable AI system rather than a standalone model, with modular components for planning, control, evaluation, and scenario-driven testing.

Real-Time Conveyor-Belt Object Detection and Robotic Sorting

Jan 2026 - May 2026

- Built a real-time AI perception and control pipeline for robotic sorting, using YOLO, OpenCV, homography-based coordinate transforms, IK, and delta-arm motion control to connect visual detections to physical actions.
- Developed the end-to-end system in Python across distributed hardware and software components, integrating edge compute, cloud vision inference, robot control, and timing logic into a closed-loop automation workflow.
- Designed post-detection refinement and motion-compensation logic to improve coordinate accuracy, account for conveyor motion, and reduce perception-to-actuation lag in deployment conditions.
- Achieved up to 99% classification accuracy and approximately 50% end-to-end physical sorting accuracy, using system-level evaluation to identify latency, conveyor-speed variation, and mechanical compliance as the main operational bottlenecks.

Experience

Field Service Engineer I

Santa Clara, CA / Ulvac Technologies, Oct 2024 - Aug 2025

- Rebuilt and installed an upgraded 8-inch etch chuck assembly, integrated a new VAT system, circuit breaker, valve-control updates, and wiring changes to improve tool flexibility, reducing unplanned downtime by 15%, and increasing wafer temperature uniformity.
- Sustained uptime across semiconductor tools across 6 high-volume fabs through preventive maintenance, calibration, and data-driven repairs, increasing uptime by 18%.
- Executed 30+ hardware and software retrofit projects across vacuum, RF, motion-handling, PLC, and HMI subsystems, improving process stability by 22% and extending tool life.
- Led root-cause analysis with engineering and manufacturing teams, implemented corrective actions, and standardized troubleshooting procedures, reducing average fault diagnosis time by 35%.